Study Session 5 — Feature Importance & Model Refinement

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Prep

- ☐ Open 05_Feature_Importance_and_Model_Refinement.ipynb.
- ☐ Load DataFrame df with: bill_total_usd, tip_usd, party_size (and dummies if available).
- ☐ Create train/test split and baseline LinearRegression.

Model & Metrics

- ☐ Fit baseline multiple regression (bill_total_usd, party_size, + optional dummies).
- ☐ Print metrics: R^2, Adjusted R^2, RMSE (train & test).
- ☐ Log quick notes: Where does the model under/over-predict?

Feature Importance

- ☐ Standardize X and fit LinearRegression on standardized features.
- ☐ Plot Standardized Coefficients (β^*) bar chart and interpret magnitudes/signs.
- ☐ Save figure via save_and_log_plot() to reports/figures/.

Multicollinearity (Optional but recommended)

- ☐ Compute VIF for predictors (drop constant in plot).
- ☐ Visualize VIF bar chart with thresholds at 5 and 10.
- ☐ Decide whether to remove/transform overlapping variables.

Diagnostics

- ☐ Plot Residuals vs Predicted with horizontal line at 0 (look for randomness).
- ☐ Check residual normality (histogram + Q–Q plot).
- ☐ If heteroscedasticity appears, try log/Box-Cox of target or robust regression.

Regularization (Stretch Goal)

- ☐ Compare Ridge and Lasso with cross-validation.
- ☐ Plot coefficient magnitudes vs α (regularization strength).
- ☐ Pick a simpler model that keeps accuracy while shrinking noise.

Wrap-Up

- ☐ Write a 3–5 sentence takeaway: which features matter most and why.
- ☐ Push updated notebook and figures to Git (meaningful commit message).
- ☐ Create a TODO for tomorrow (e.g., interactions or nonlinearity).

Notes